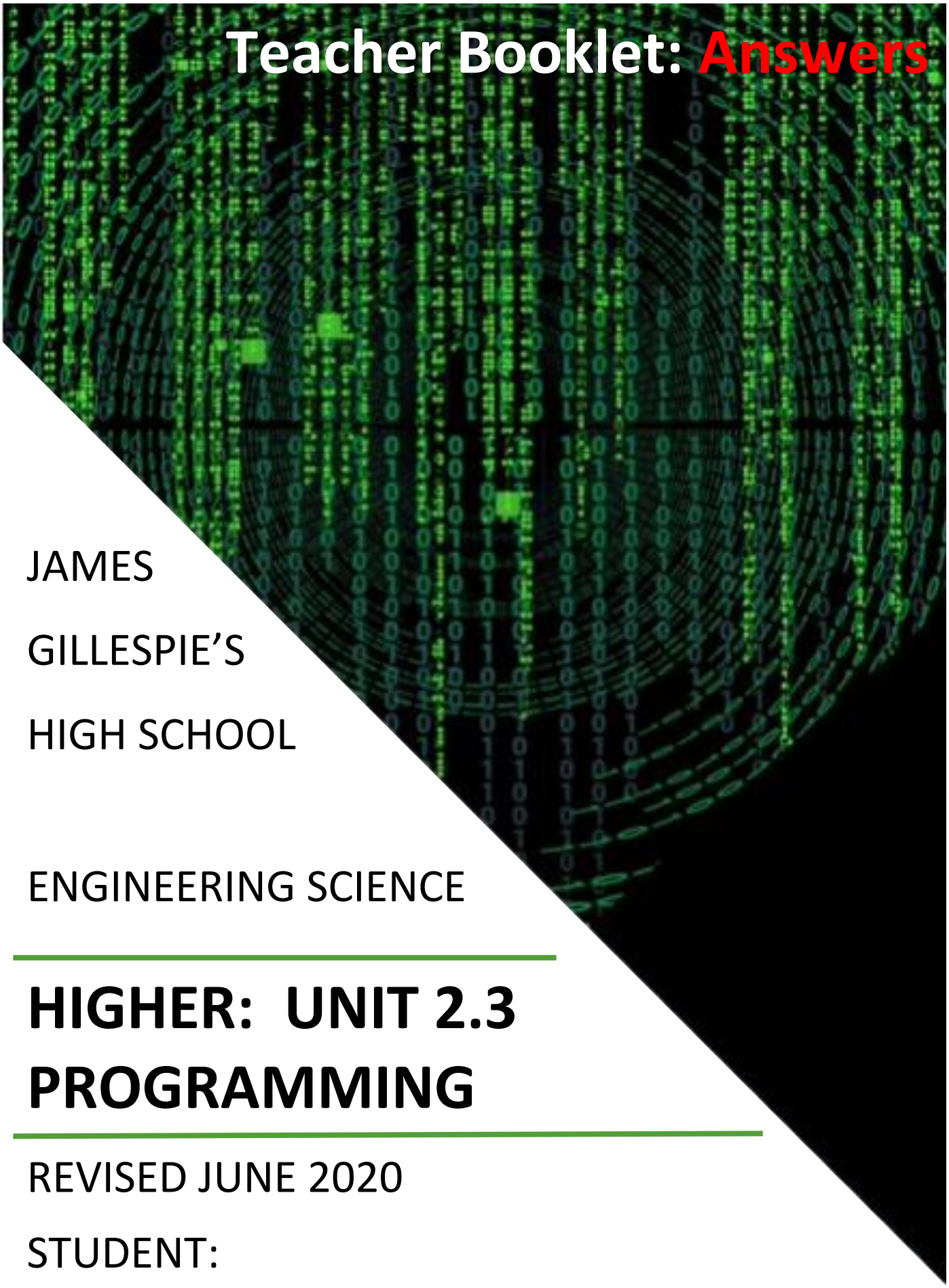


Teacher Booklet: Answers

JAMES

GILLESPIE'S

HIGH SCHOOL

ENGINEERING SCIENCE

HIGHER: UNIT 2.3 PROGRAMMING

REVISED JUNE 2020

STUDENT:

YEAR: TEACHER:

Important: All flowcharts and PBASIC example answers given in this booklet are not definitive. Students may still obtain correct answers and full marks, while giving different answers.

All answers must show clear understanding and should conform to the current SQA standards. Please check the standard in SQA documentation to give students up to date advice on answers.

1

a) let dirs = %00000011
b) let dirs = %11110000
c) let dirs = %10101010
d) let dirs = %11000000
e) let dirs = %01101000

2

a) Any three from the following:

- Relatively inexpensive to buy, much cheaper than discrete digital logic components and wiring them.
- Very small and this can reduce the size of the commercial product.
- Can be connected to the internet, so they can be interfaced with from anywhere in the world.
- Can be reprogrammed, where discrete electronic components may have to be redesigned.
- increased reliability and reduced stock inventory (as one microcontroller replaces several parts).
- simplified product assembly and smaller end products.
- greater product flexibility and adaptability since features are programmed into the microcontroller and not built into the electronic hardware.
- rapid product changes or development by changing the program and not the electronic hardware.

Extra info: Disadvantages:

i. Internet connected microcontrollers can be vulnerable (could be hacked).
ii. Engineer must know how to code to use them.
iii. Vulnerable to static shocks and voltage spikes, so can act up.
iv. There's a lot of choice, so engineer must be sure they choose the optimal one.

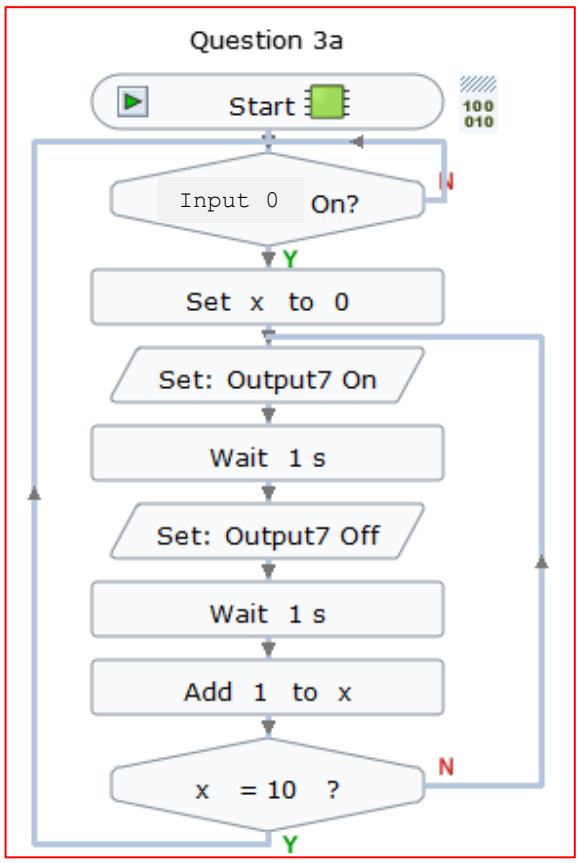
b)

- RAM: Random Access Memory.
  Is a temporary storage area. Used to name IO ports, remember numbers of data-log information coming in. All data is lost when the power goes off.
- ROM: Read Only Memory.
  Contains a base set of instructions that allow the microcontroller to function. The microcontroller will read this memory when it first switches on and will then know how to read the user-made programme.
- EEPROM: Electronically Erasable Programmable Read Only Memory.
  Is where the user programme is stored. This cannot be over written by the ALU but by connecting the microcontroller to another computer, the user can delete or amend the programme.

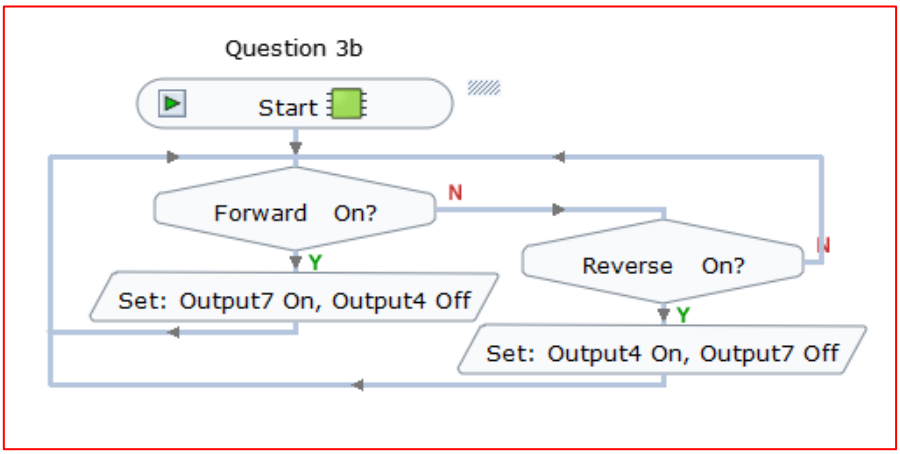
3

Important: Students who studied N5 should know how to do these questions. These are example answers and are NOT the only valid solutions. Students must use SQA standards in answers in their jotters – check SQA documentation for latest standards of accepted answers and formats.

a) Answer must show
   a. Input decision box
   b. Output 7 flashing on/off
   c. Loop 10 times
   d. Return to beginning for looped programme

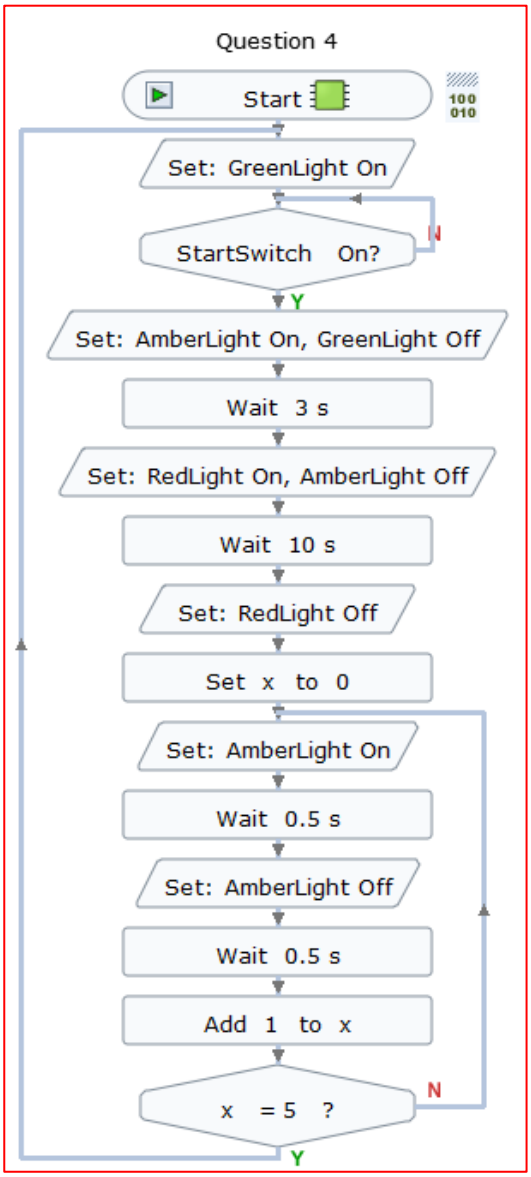


b) Answer must show
   a. 2 decision boxes testing state of inputs
   b. Motor being switched on/off appropriately

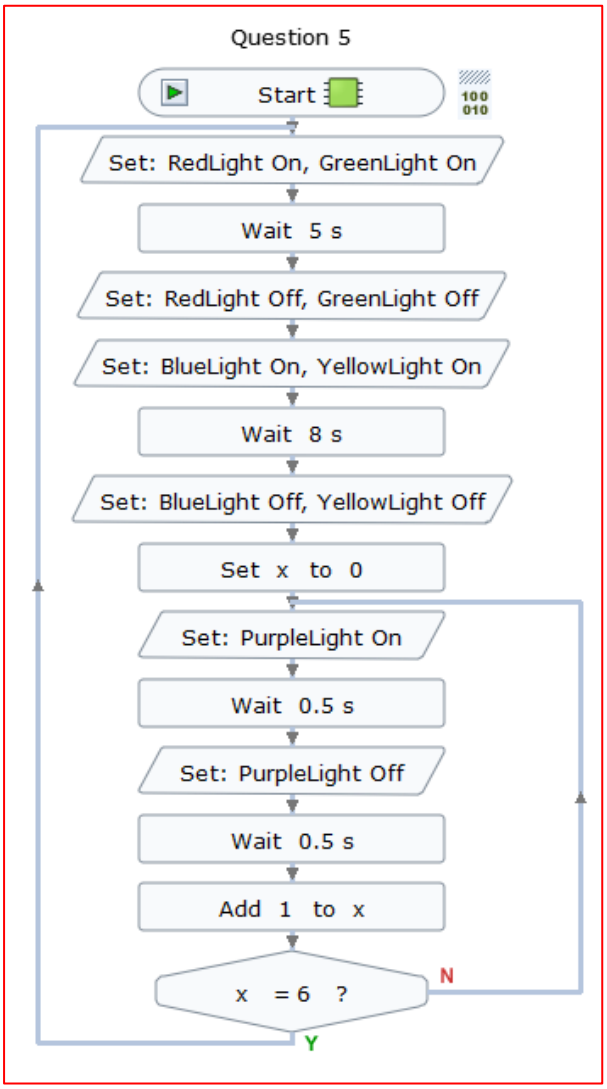


4

Answer must show correct sequence of lights, as shown in states image. Timings should be suitable, but don't have to match this model answer.

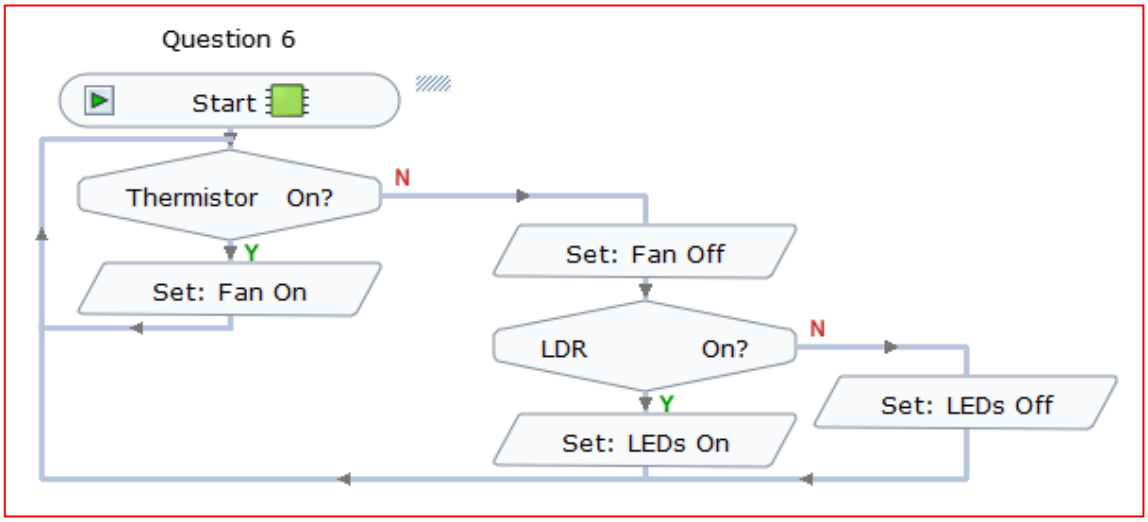


5



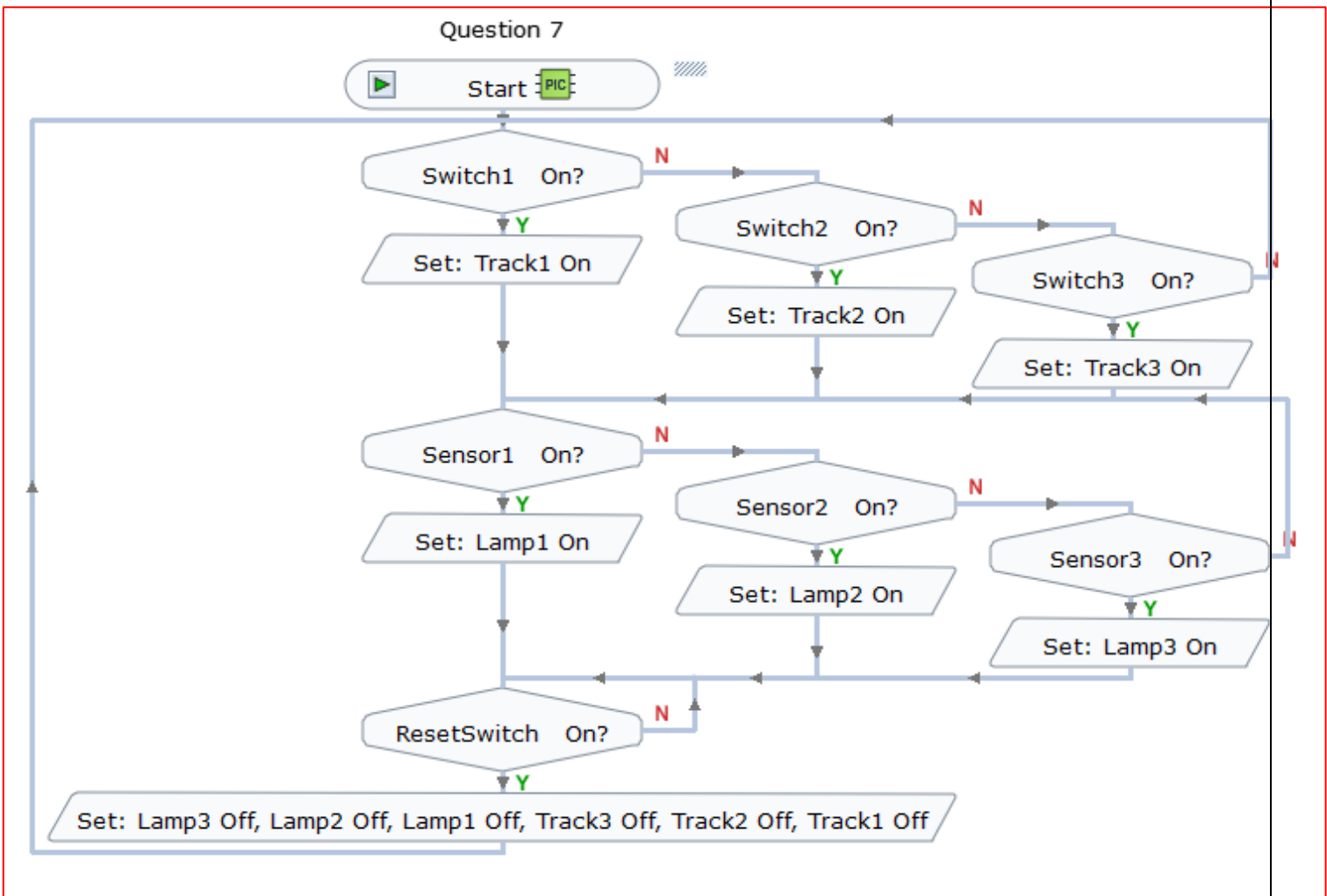
6

Key: Students must demonstrate understanding that fan turns on/off based on thermistor high/low signal to microcontroller AND LED bank on/off according to LDR high/low signal to microcontroller. No need to create analogue circuitry, but students may find this helpful for their processing and could be set as homework portion of Q.



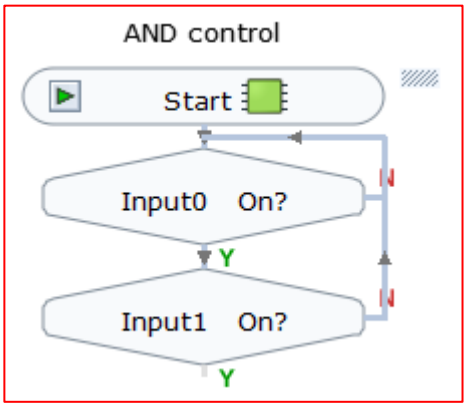
7

Key: Students must demonstrate understanding of sequencing, multiple inputs and outputs according to switches and sensors AND must turn off outputs before looping programme. Each student's understanding of this system will be different, as long as they can explain their programme reasonable, it will be an acceptable answer.

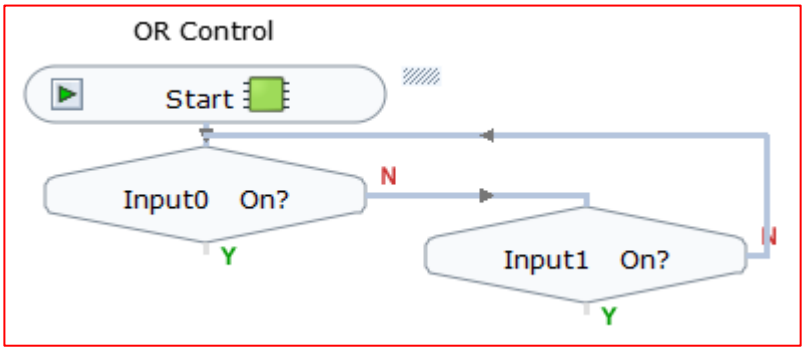


8

a)


AND control
Start
Input0 On?
N
Y
Input1 On?
N
Y

b)


OR Control
Start
Input0 On?
N
Y
Input1 On?
N
Y

9

```
let dirs = %11100000                ‘Setting up the directory. Set pin 7, 6, 5 as
                                    output
MAIN:                               ‘Make a label called ‘MAIN’

MAIN:

            high 7                  ‘switch pin 7 on
            pause 10000             ‘wait for 10 seconds
            high 6                  ‘switch pin 6 on
            pause 2000              ‘wait for 2 seconds
            low 7                   ‘switch pin 7 off
            low 6                   ‘switch pin 6 off
            high 5                  ‘switch pin 5 on
            pause 10000             ‘wait for 10 seconds
            low 5                   ‘switch pin 5 off
            high 6                  ‘switch pin 6 on
            pause 2000              ‘wait for 2 seconds
            low 6                   ‘switch pin 6 off
            end OR goto MAIN        ‘ends program OR returns to MAIN
```

OR PREFERRED ANSWER:

```
let dirs = %11100000                ‘Setting up the directory.  set pin 7, 6, 5 as
                                    output

MAIN:                               Make a label called ‘MAIN’

symbol red = 7                      'Define pin 7 ‘red’
symbol amber = 6                    'Define pin 6 ‘amber’
symbol green = 5                    'Define pin 5 ‘green’

MAIN:                               ‘Make a label called ‘MAIN’

            high red                ‘switch pin 7 (red) on
            pause 10000             ‘wait for 10s
            high amber              ‘switch pin 6 (amber) on
            pause 2000              ‘wait for 2s
            low red                 ‘switch pin 7 (red) off
            low amber               ‘switch pin 6 (amber) off
            high green              ‘switch pin 5 (green) on
            pause 10000             ‘wait for 10s
            low green               ‘switch pin 5 (green) off
            high amber              ‘switch pin 6 (amber) on
            pause 2000              ‘wait for 2s
            low amber               ‘switch pin 6 (amber) off
            end OR goto MAIN        ‘ends program OR returns to MAIN
```

10 Important: these are not difinitive answers. There are many ways to write a programme that means the students have correct answers, however these are ideal model answers. Student will receive marks for different answers, as long as the syntax is correct. Students must comment each line to demonstrate understanding.

Q3a:

```
let dirs = %11110000                    'setting up the directory.  Set 0-3 as inputs and 4-7 as
                                        outputs
symbol counter = b0                     'define the variable 'counter'

MAIN:                                   'make a label called 'MAIN'
          if pin0=1 then FLASH          'if input0 is high then goto FLASH subprogramme
          goto MAIN                     'return to MAIN programme

FLASH:
          for counter 1 to 7            'start a for...next loop
          high 7                        'switch pin 7 high
          pause 1000                    'wait 1 second
          low 7                         'switch pin 7 low
          pause 1000                    'wait 1 second
          next counter                  'end of for...next loop
          goto MAIN                     'return to MAIN programme
```

Q3b: Students should also include low 7 and low 4 at appropriate points according to their programme

```
let dirs = %11110000                    'setting up the directory.  Set 0-3 as inputs and 4-7 as
                                        outputs
symbol forward = 0                      'define pin 0 with the name 'forward'
symbol reverse = 2                      'define pin 2 with the name 'reverse'

MAIN:                                   'make a label called 'main'
          if forward=1 then high 7      'if input0 is high then switch pin 7 high
          if reverse=1 then high 4      'if input2 is high then switch pin 4 high
          goto MAIN                     'return to MAIN programme
```

Q4:

```
let dirs = %11110000                    'Setting up the directory.  Set 0-3 as inputs and 4-7 as
                                        outputs
symbol counter = b0                     'define the variable 'counter'
symbol start_switch = 3                 'define pin 3 with the name 'start_switch'
symbol red = 7                          'define pin 7 with the name 'red'
symbol amber = 6                        'define pin 6 with the name 'amber'
symbol green = 5                        'define pin 5 with the name 'green'

MAIN:                                           'make a label called 'main'
          high green                            'switch pin 5 high
          if start_switch=1 then high amber     'if input3 is high then switch pin 6 high
          low green                             'switch pin 5 low
          pause 3000                            'wait 3s
          high red                              'switch pin 7 high
          low amber                             'switch pin 6 low
          pause 10000                           'wait 10s
          low red                               'switch pin 7 low
          for counter = 1 to 5                  'start a for...next loop
          high amber                            'switch pin 6 high
          pause 500                             'wait 0.5s
          low amber                             'switch pin 6 low
          pause 500                             'wait 0.5s
          next counter                          'end of for...next loop
          goto MAIN                             'return to MAIN label
```

Q5:

```
let dirs = %00000000                  'setting up the directry. Set pin 0-7 as outputs
symbol counter = b0                   'define the variable 'counter'
symbol yellow = 6                     'define pin 6 with the name 'yellow'
symbol purple = 5                     'define pin 5 with the name 'purple'
symbol blue = 3                       'define pin 3 with the name 'blue'
symbol green = 2                      'define pin 2 with the name 'green'
symbol red = 0                        'define pin 0 with the name 'red'

MAIN:                                           'make a label called 'MAIN'
         high red                               'switch pin 0 high
         high green                             'switch pin 2 high
         pause 5000                             'wait 5 seconds
         low red                                'switch pin 0 low
         low green                              'switch pin 2 low
         high blue                              'switch pin 3 high
         high yellow                            'switch pin 6 high
         pause 8000                             'wait 8 seconds
         low blue                               'switch pin 3 low
         low yellow                             'switch pin 6 low
         for counter = 1 to 6 high purple       'start a for...next loop
         pause 500                              'wait 0.5 seconds
         low purple                             'switch pin 5 low
         pause 500                              'wait 0.5 seconds
         next counter                           'end for...next loop
         goto MAIN                              'return to MAIN label
```

OR another valid answer example:

```
let dirs = %00000000          'setting up the directory.  Set pin 0-7 as outputs
symbol counter = b0           'define the variable 'counter'
symbol yellow = 6             'define pin 6 with the name 'yellow'
symbol purple = 5             'define pin 5 with the name 'purple'
symbol blue = 3               'define pin 3 with the name 'blue'
symbol green = 2              'define pin 2 with the name 'green'
symbol red = 0                'define pin 0 with the name 'red'

MAIN:                                   'make a label called 'MAIN'
         let pins = %00000101           'switch pin 0 & 2 high
         pause 5000                     'wait 5 seconds
         let pins = %01001000           'switch pin 0 & 2 low, 3 & 6 high
         pause 8000                     'wait 8 seconds
         let pins = %00000000           'switch pin 3 & 6 low
         for counter = 1 to 6           'start a for...next loop
         let pins = %00100000           'switch pin 5 high
         pause 500                      'wait 0.5 seconds
         let pins = %00000000           'switch pin 5 low
         pause 500                      'wait 0.5 seconds
         next counter                   'end for...next loop
         goto MAIN                      'return to MAIN label
```

Q6: Students will need to define their own input and output pins for this question. As such the pin assignments in this example are not definitive.

```
let dirs = %11110000          'setting up the directory. S  et 0-3 as inputs and 4-7 as
                              outputs
symbol thermistor = 0         'define pin 0 with the name 'thermistor'
symbol LDR = 2                'define pin 2 with the name 'LDR'
symbol fan = 4                'define pin 4 with the name 'fan'
symbol LEDs = 6               'define pin 6 with the name 'LEDs'

MAIN:                                     'make a label called "MAIN"
        if thermistor = 1 then high fan   'check pin 0 input and turn pin 4 high
        else low fan                      'else pin 4 low
        if LDR = 1 then high LEDs         'if pin 2 is high, turn pin 6 high
        else low LEDs                     'else pin 6 low
        goto MAIN                         'return to MAIN label
```

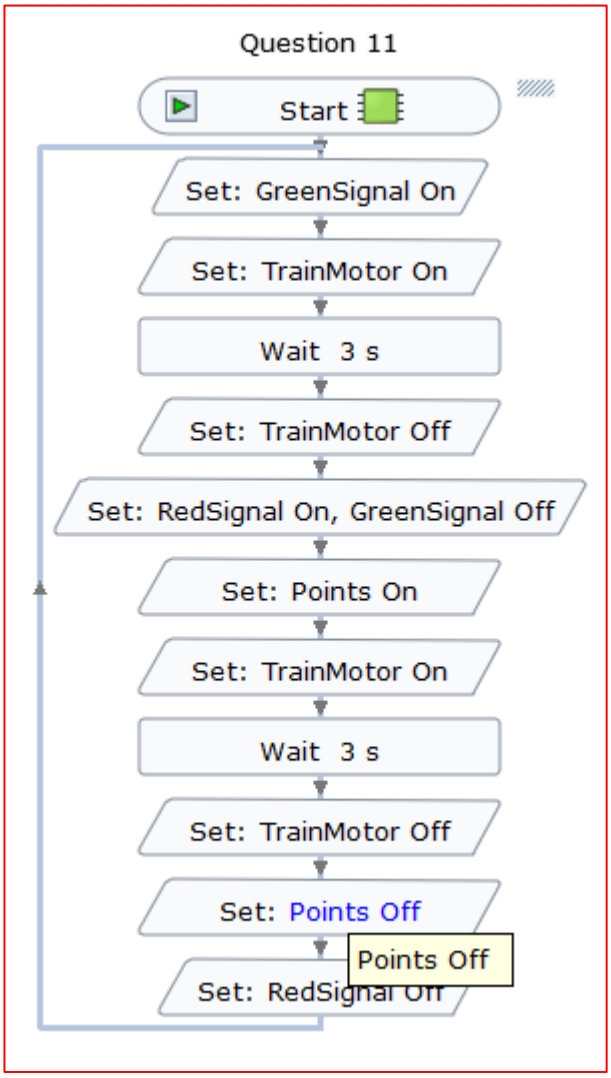
Q7: Students can assume they are using a generic 14 pin chip

```
let dirs = %11111110000000       'Setting up the directory. Set 0-6 as inputs
                                 and 7-14 as outputs

symbol switch1 = 0          'define pin 0 with the name 'switch1'
symbol switch2 = 1          'define pin 1 with the name 'switch2'
symbol switch3 = 2          'define pin 2 with the name 'switch3'
symbol sensor1 = 3          'define pin 3 with the name 'sensor1'
symbol sensor2 = 4          'define pin 4 with the name 'sensor2'
symbol sensor3 = 5          'define pin 5 with the name 'sensor3'
symbol reset_switch = 6     'define pin 6 with the name 'reset_switch'
symbol track1 = 7           'define pin 7 with the name 'track1'
symbol track2 = 8           'define pin 8 with the name 'track2'
symbol track3 = 9           'define pin 9 with the name 'track3'
symbol lamp1 = 10           'define pin 10 with the name 'lamp1'
symbol lamp2 = 11           'define pin 11 with the name 'lamp2'
symbol lamp3 = 12           'define pin 12 with the name 'lamp3'

MAIN:                                     'make a label called "MAIN"
  if switch1 = 1 then high track1         'if pin 0 is high, turn pin 6 high
  if switch2 = 1 then high track2         'if pin 1 is high, turn pin 7 high
  if switch3 = 1 then high track3         'if pin 2 is high, turn pin 8 high
  else main                               'else return to main programme
  if sensor1 = 1 then high lamp1          'if pin 3 is high, turn pin 10 high
  if sensor2 = 1 then high lamp2          'if pin 4 is high, turn pin 11 high
  if sensor3 = 1 then high lamp3          'if pin 5 is high, turn pin 12 high
  if reset_switch = 1 then let pins = %00000000000000
                                          'reset switch turns off all outputs
  goto MAIN                               'return to main label
```

11

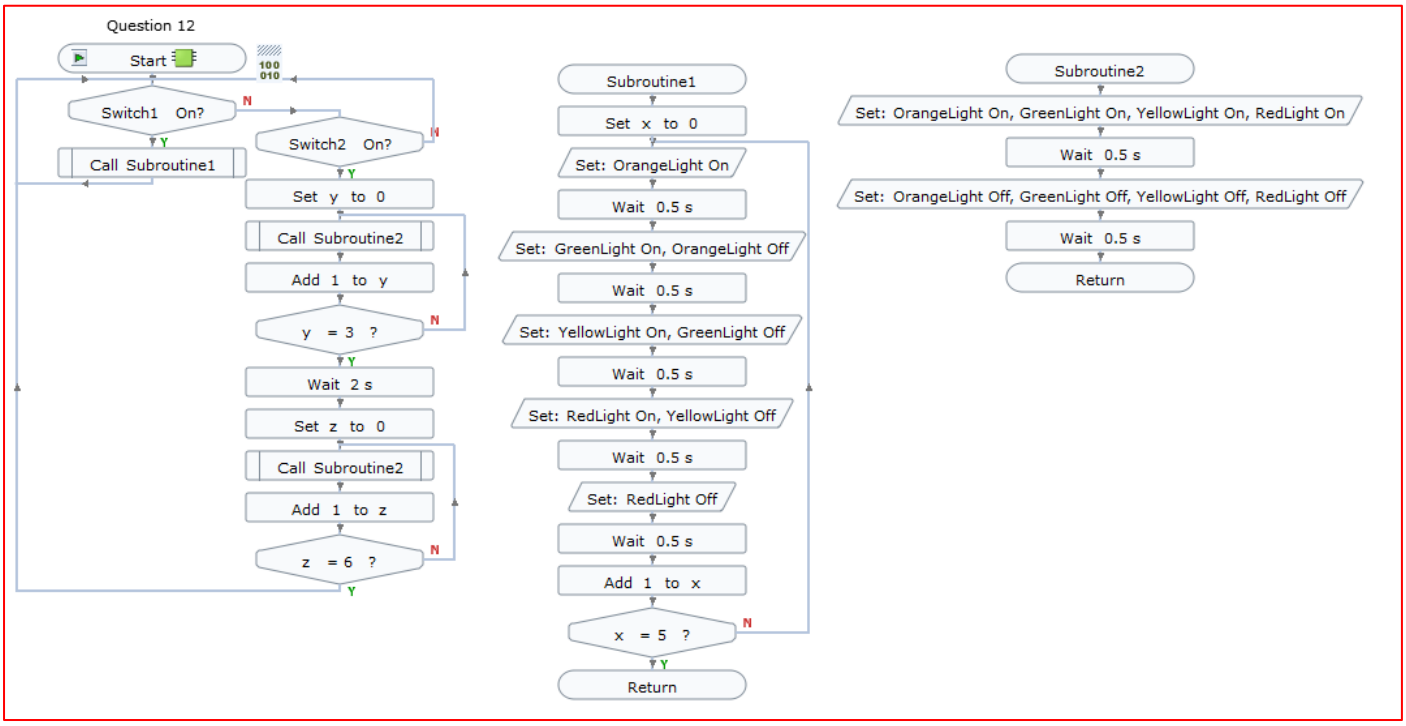



```
let dirs = %11110000      'Setting up the directory.  Set 0-3 as inputs
                          and 4-7 as outputs
symbol train_motor = 7    'define pin 7 with the name 'train_motor'
symbol points = 6         'define pin 6 with the name 'points'
symbol green_signal = 5   'define pin 5 with the name 'green_signal'
symbol red_signal= 4      'define pin 4 with the name 'red_signal'

MAIN:                     'make a label called "MAIN"
     high green_signal    'turn pin 5 high
     high train_motor     'turn pin 7 high
     pause 3000           'wait 3 seconds
     low train_motor      'turn pin 7 low
     low green_signal     'turn pin 5 low
     high red_signal      'turn pin 4 high
     high points          'turn pin 6 high
     high train_motor     'turn pin 7 high
     pause 3000           'wait 3 seconds
     low train_motor      'turn pin 7 low
     low points           'turn pin 6 low
     low red_signal       'turn pin 4 low
     goto MAIN            'return to MAIN label
```

12

Example flowchart: Students must use subprocedures, but could put loop in subprocedure or main program, as shown. Preference for loop in main programme.



PBASIC: Students must use the following symbol assignments for correct marks, as these are given in the question. Sub-procedure code can be writen different ways but must accurately follow the flowchart, which is why 2 different examples are given in this answer. Efficiency and style can be discussed with students when showing this answer. Note PBASIC is spread over 2 pages.

```
let dirs = %11110000          'setting up the directory. Set 0-3 as inputs and 4-7 as
                              outputs
symbol switch1 = 1            'define pin 1 with the name 'switch1'
symbol switch2 = 2            'define pin 2 with the name 'switch2'
symbol red = 7                'define pin 7 with the name 'red'
symbol yellow = 6             'define pin 6 with the name 'yellow'
symbol green = 5              'define pin 5 with the name 'green'
symbol orange = 4             'define pin 4 with the name 'orange'
symbol counter1 = b0          'define variable 'counter1'
symbol counter2 = b1          'define variable 'counter2'
symbol counter3 = b2          'define variable 'counter3'

MAIN:                                                 'make a label called "MAIN"
  if switch1 = 1 then gosub SUBROUTINE1               'when switch1 is high call subroutine1
  if switch2 = 1 then for counter1 = 1 to 3           'when switch2 is high start loop
  gosub SUBROUTINE2                                   'call subroutine2
  next counter1                                       'complete loop
  pause 2000                                          'wait 2 seconds
  for counter2 = 1 to 6                               'begin new loop
  gosub SUBROUTINE2                                   'call subroutine2
  next counter2                                       'complete loop
  else goto MAIN                                      'return to MAIN
```

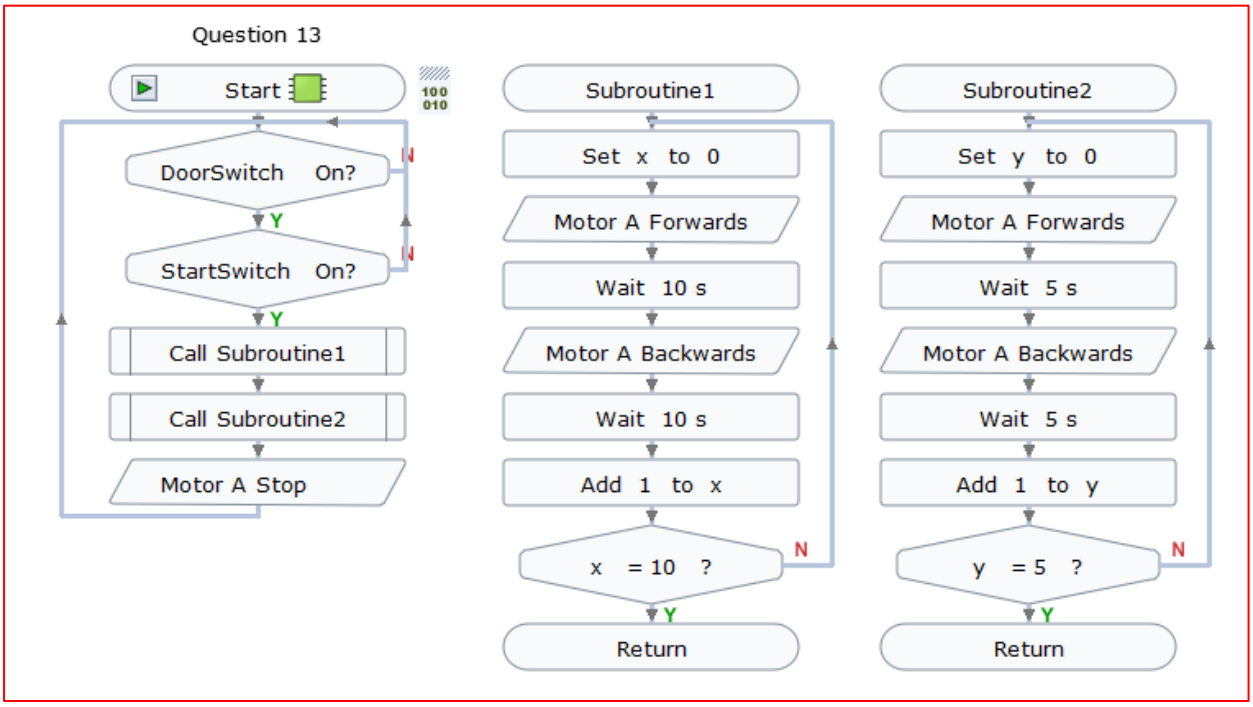
```
SUBROUTINE1:                        'make a label called "SUBROUTINE1"

      for counter3 = 1 to 5         'start a new loop
      high orange                   'switch pin 4 high
      pause 500                     'wait 0.5 seconds
      low orange                    'switch pin 4 low
      high green                    'switch pin 5 high
      pause 500                     'wait 0.5 seconds
      low green                     'switch pin 5 low
      high yellow                   'switch pin 6 high
      pause 500                     'wait 0.5 seconds
      low yellow                    'switch pin 6 low
      high red                      'switch pin 7 high
      pause 500                     'wait 0.5 seconds
      low red                       'switch pin 7 low
      next counter3                 'complete loop
return                              'return from subprocedure

SUBROUTINE2                         'make a label called "SUBROUTINE2"
      high orange                   'switch pin 4 high
      high green                    'switch pin 5 high
      high yellow                   'switch pin 6 high
      high red                      'switch pin 7 high
      pause 500                     'wait 0.5 seconds
      low orange                    'switch pin 4 low
      low green                     'switch pin 5 low
      low yellow                    'switch pin 6 low
      low red                       'switch pin 7 low
      pause 500                     'wait 0.5 seconds
return                              'return from subprocedure
```

13

Example flowchart: Students must use subprocedures



PBASIC:

```
let dirs = %11110000        'setting up the directory. Set 0-3 as inputs and 4-7
                            as outputs

symbol door_switch = 1      'define pin 1 with the name 'door_switch'
symbol start_switch = 2     'define pin 2 with the name 'start_switch'
symbol motor_forwards = 7   'define pin 7 with the name 'motor_forwards'
symbol motor_backwards = 6  'define pin 6 with the name 'motor_backwards'
symbol counter1 = b0        'define variable 'counter1'
symbol counter2 = b1        'define variable 'counter2'

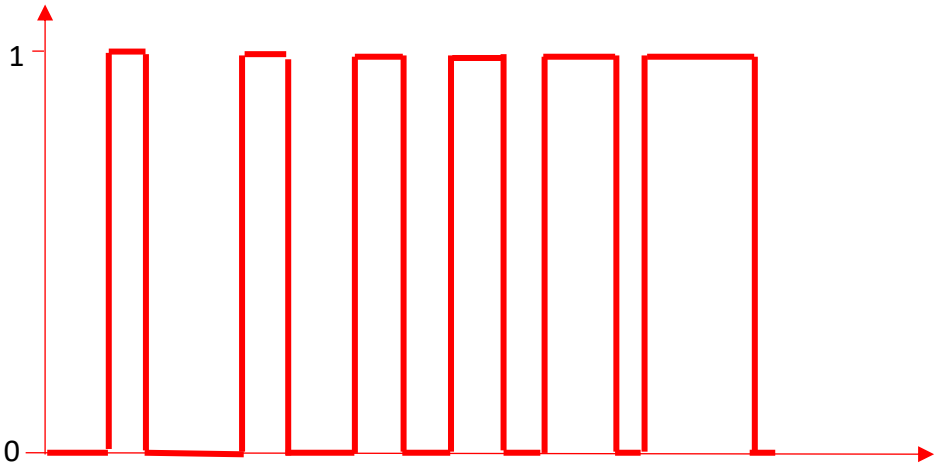
MAIN:                                       'make a label called "MAIN"
     if door_switch = 1 then                'when pin 1 is high continue
          elseif start_switch = 1 then      'when pin 2 is high continue
          gosub subroutine1                 'call subroutine1
          gosub subroutine2                 'call subroutine2
     else goto main                         'return to main

SUBROUNTINE1                            'make a label called "SUBROUNTINE1"
     for counter = 1 to 10          'start a new loop
     high motor_forwards            'switch pin 7 high
     pause 10000                    'wait 10 seconds
     low motor_forwards             'switch pin 7 low
     high motor_backwards           'switch pin 6 high
     pause 10000                     10 seconds
     low motor_backwards            'switch pin 6 low
     next counter                   'complete loop
return                              'return from subprocedure

SUBROUNTINE2                            'make a label called "SUBROUNTINE2"
     for counter = 1 to 5           'start a new loop
     high motor_forwards            'switch pin 7 high
     pause 5000                     'wait 5 seconds
     low motor_forwards             'switch pin 7 low
     high motor_backwards           'switch pin 6 high
     pause 5000                     'wait 5 seconds
     low motor_backwards            'switch pin 6 low
     next counter                   'complete loop
return                              'return from subprocedure
```

14

Example answer: Important that students are accurate at showing pulses where mark size is increasing, and space size is decreasing. All marks must be same height, all spaces must be at 0.



15

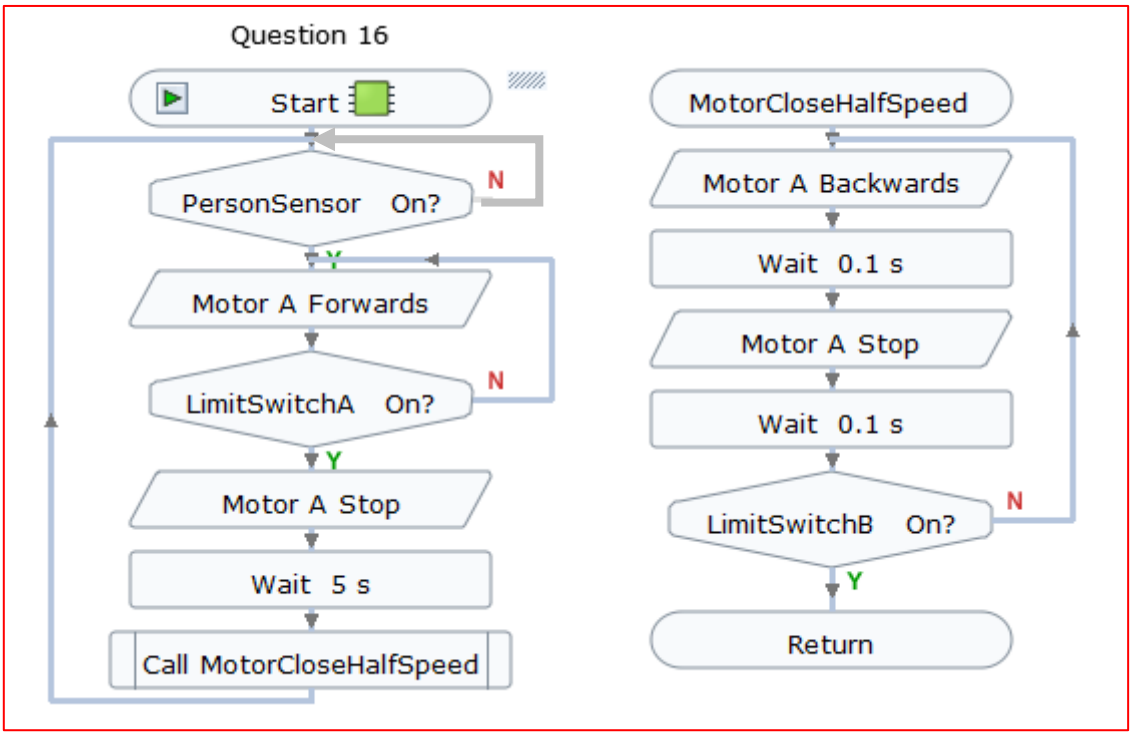
Note: this is a long question and students can be left to work independently. They will need to take time to download and understand the simulartion.
As students are to use Yenka, no example answer. Teacher can demonstrate task on the board.

https://www.yenka.com/activities/PWM_Speed_Control_of_d.c._Motor_-_Activity/

If students understanding is advance, teacher can suggest continuation of activity using:
https://www.yenka.com/activities/Pulse_Width_Modulation_III_-_Activity/

16

a) Example flowchart:



b)PBASIC:

```
let dirs = %11110000          'setting up the directory. Set 0-3 as inputs and 4-7 as
                              outputs
symbol person_sensor = 2      'define pin 2 with the name 'person_sensor'
symbol limit_switcha = 1      'define pin 1 with the name 'limit_switcha'
symbol limit_switchb = 0      'define pin 0 with the name 'limit_switchb'
symbol motor_forwards = 7     'define pin 7 with the name 'motor_forwards'
symbol motor_backwards = 6    'define pin 6 with the name 'motor_backwards'
SUBROUTINE                    'Label SUBROUTINE for motor close half speed:

MAIN:                                                 'make a label called "MAIN"
   if person_sensor = 1 then high motor_forwards      'when pin 2 is high switch pin 7 high
   else goto MAIN                                     'return to MAIN
   if limit_switcha = 1 then low motor_forwards       'when pin 1 is high switch pin 7 low
   pause 5000                                         'wait 5 seconds
   gosub motorclosehalfspeed                          'call SUBROUTINE
   else goto MAIN                                     'return to MAIN

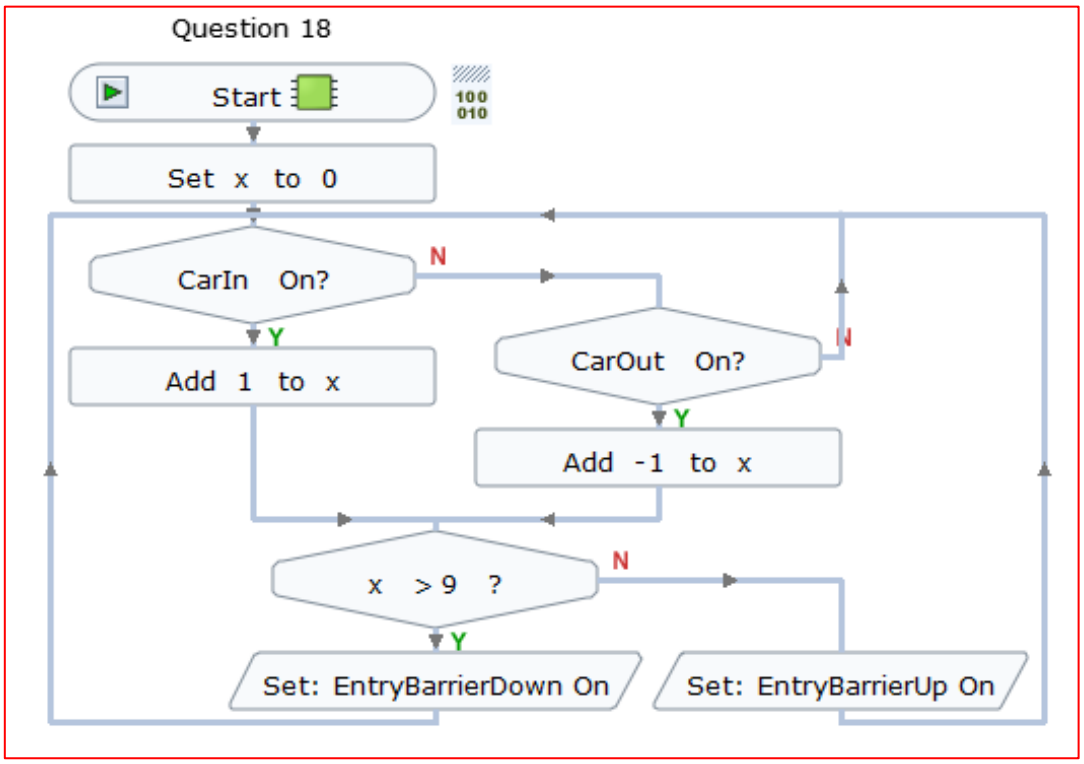
SUBROUTINE:                                           'make a label called "SUBROUTINE"
      high motor_backwards                                  'switch pin 6 high
      pause 100                                             'wait 0.1 seconds
      low motor_backwards                                   'switch pin 6 low
      pause 100                                             'wait 0.1 seconds
      if limit_switchb = 0 then goto SUBROUNTINE            'loop to SUBROUTINE
      else return                                                 'go back to MAIN programme
```

17

```
let dirs = %11110000          'setting up the directory.  Set 0-3 as inputs and 4-
                              7 as outputs
symbol itemIN = 0             'define pin 0 with the name 'itemIN'
symbol itemOUT = 3            'define pin 3 with the name 'itemOUT'
symbol warning = 7            'define pin 7 with the name 'warning'


MAIN:                                           'make a label called "MAIN"
      let x = 0                                 'define variable 'x' as 0
      if itemIN = 1 then let x = x+1            'when pin 0 is high add 1 to x
      elseif itemOUT = 1 then let x = x-1       'when pin 3 is high subtract 1 from x
      if x < 3 then high warning          'if x is less than three switch pin 7 high
      else low warning                          'if x is not less than three pin 7 low
      else goto MAIN                            'return to MAIN
```

18 The example flowchart below is oversimplified and does not consider the need for a limit on the raising/lowering of the barrier. This is an acceptable answer if the engineer assumes the barrier has an inbuilt timer/system to stop the motor driving the barrier beyond its limit, however this is unlikely. If a student produces a flowchart like this, discuss the issues of this oversimplification with them.

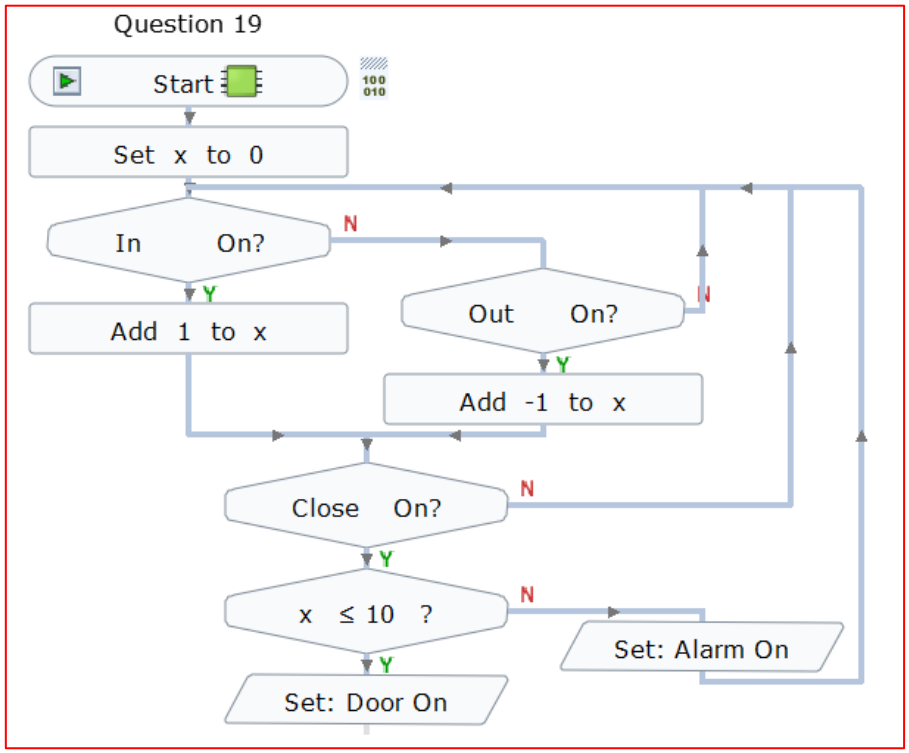



```
let dirs = %11110000          'Setting up the directory. Set 0-3 as inputs and 4-7 as
                              outputs

symbol CarIn= 0               'define pin 0 with the name 'CarIn'
symbol CarOut = 1             'define pin 1 with the name 'CarOut'
symbol EntryBarrierDown = 7   'define pin 7 with the name 'Entry Barrier Down'
symbol EntryBarrierUp = 6     'define pin 6 with the name 'Entry Barrier Up'

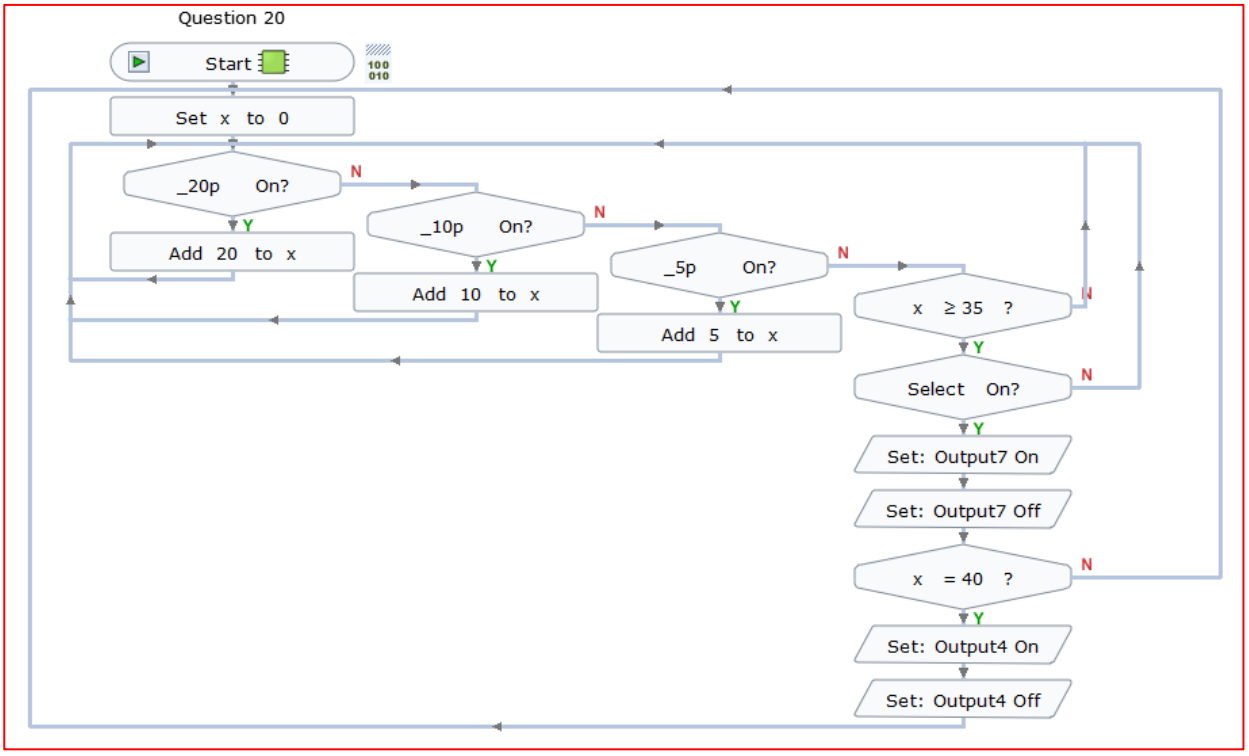
MAIN:                                          'make a label called "MAIN"
      let x = 0                                'define variable 'x' as 0
      if CarIn = 1 then let x = x+1            'when pin 1 is high add 1 to x
            elseif CarOut = 1 then let x = x-1 'when pin 3 is high subtract 1 from x
            if x > 9 then high EntryBarrierDown 'if x is less than three switch pin 7 high
            else high EntryBarrierUp           'if x is not less than three pin 7 low
      else goto MAIN                           'return to main
```

19



Flowchart is incomplete as no clear instructions on what happens in programme once door closes (and it is illogical that that would end the programme). However, students may give any reasonable answer here. PBASIC will be similar to Q18 but with changed variables.

20



Students must not just copy the PBASIC from the yenka editor – it does not follow the current SQA requirements. However they can use it to help them make sense of the requirements of their programme if they find that helpful.

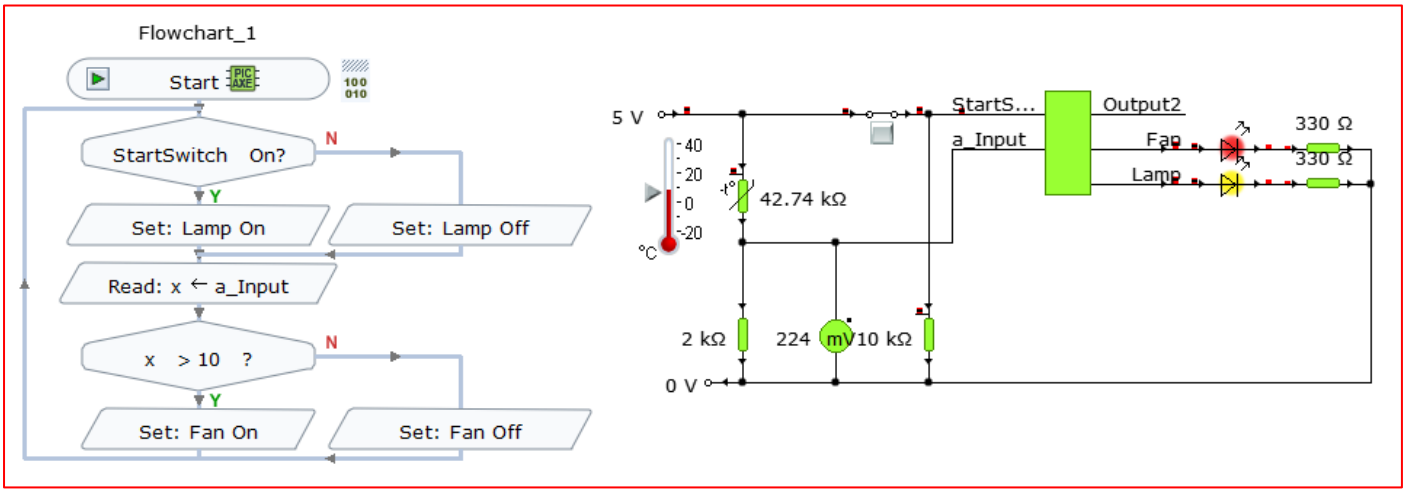
PBASIC:

```
let dirs = %11110000        'Setting up the directory. Set 0-3 as inputs and 4-7
                            as outputs

symbol _20p = 0             'define pin 0 with the name '_20p'
symbol _10p = 1             'define pin 1 with the name '_10p'
symbol _5p = 2              'define pin 2 with the name '_5p'
symbol select = 3           'define pin 3 with the name 'select'

MAIN:                                       'make a label called "MAIN"
     let x = 0                              'define variable 'x' as 0
     if _20p = 1 then let x = x+20          'when pin 0 is high add 20 to x
     elseif _10p = 1 then let x = x+10      'when pin 1 is high add 10 to x
     elseif _5p = 1 then let x = x+5        'when pin 2 is high add 10 to x
          if x ≥ 35 then high 7             'if x is greater than or equal to
                                            35 switch pin 7 high
          low 7                             'switch pin 7 low
          elseif x = 40 then high 4         'if x equals 40 switch pin 4 high
          low 4                             'switch pin 7 low
     else goto MAIN                         'return to main
```

21 Students' answers must show an understanding of designing with a voltage divider circuit, and have appropriate resistor values, allowing the microcontroller to function when the temperatures is too high. This can be modelled with a temperature sensor, however students should be confident in designing a voltage divider circuit by this point. In the example below the red LED indicated when the fan would be on (set to come on when temperature is above 15°C). Flowcharts without working circuit designs are not acceptable at this level. Students must use the analogue input, and appropriate flowcharting to assign the analogue input to a variable which is then used to determine whether the fan is on or off.



PBASIC: Students answer must include the line "readACD 0,x" to receive full marks.

```
let dirs = %00000111          'Settling up the directory.  Set 0-2 as outputs and 3-7
                              as outputs

symbol startSwitch = 5        'define pin 5 with the name 'startSwitch'
symbol lamp = 0               'define pin 0 with the name 'lamp'
symbol fan = 1                'define pin 1 with the name 'fan'


MAIN:                                       'make a label called "MAIN"
      if startSwitch = 1 then high lamp     'if start switch is pressed, lamp turns on
      else low lamp                         'otherwise lamp off
      readADC 4,x                           'read analogue input at pin 4 and assign to
                                            variable x
      if x > 10 then high 1                 'if x is greater than 10 turn on the fan
      else low 1                            'else turn fan off
      goto MAIN                             'return to main programme
```

Exam Style Questions

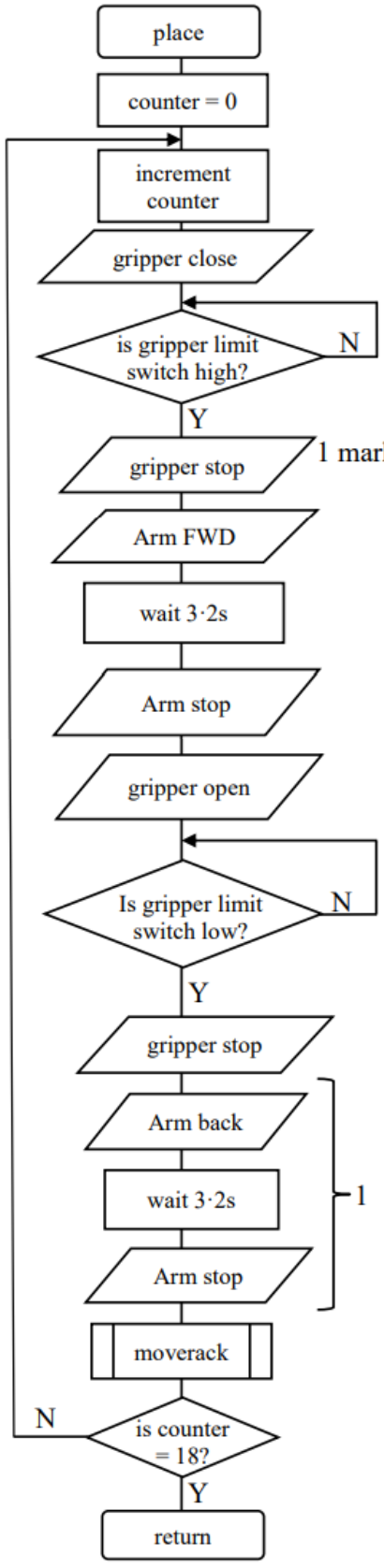
Q1:

Question			*Mark Allocation*			*Marks*	
1	**a**	**i**	feedspeed:	if pin 1 = 1 then finefeed		2	
				if pin 2 = 1 then draftfeed		2	
				return			
			finefeed:	gosub fine		1	
				return			
			draftfeed:	gosub draft		1	
				return	1 for all 3 returns	1	**7**
		ii	draft:	for b0 = 1 to 255		1	
				high 7			
				pause 10			
				low 7	1 high & low 7	1	
				pause 2	1 both pauses	1	
				next b0		1	
				return	1 including label	1	**5**

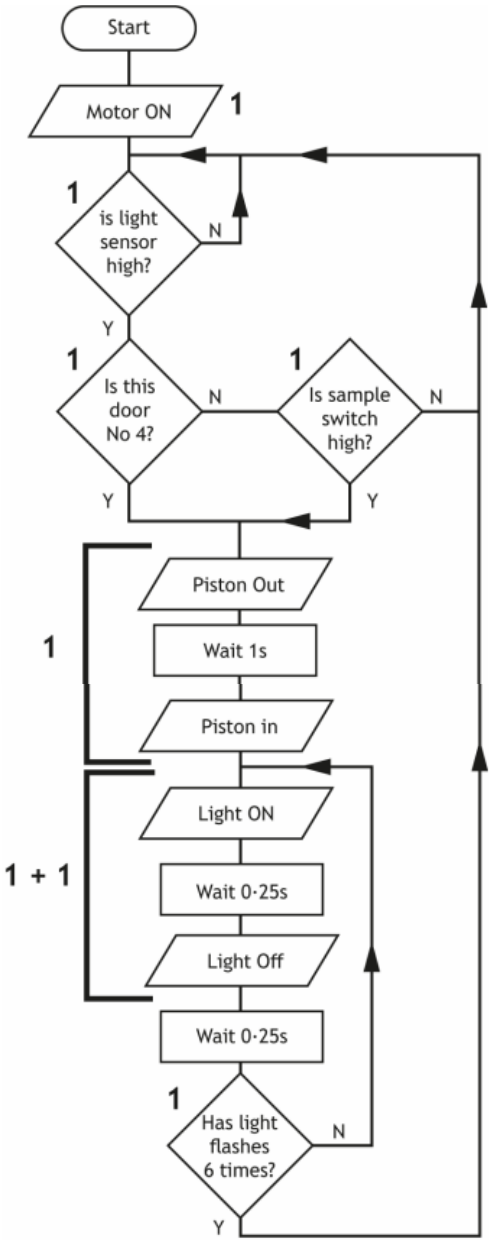
Q2:

5.		main:	if pin0 = 0 then delay	1 mark condition; 1 destination		
		test1:	if pin5 = 0 then main	1 mark condition; 1 destination		
			gosub wiper	1 mark for all four gosub wiper		
			goto main	1 mark for both goto main		
		delay:	if pin1 = 0 then wash	1 mark condition; 1 destination		
			pause 1500	1 mark		
			gosub wiper	(mark awarded above)		
			goto delay	1 mark		
		wash:	if pin 2 = 0 then main	1 mark condition; 1 destination		
			high 6	1 mark		
		again:	gosub wiper	(mark awarded above)		
			if pin 2 = 1 then again	1 mark condition; 1 destination		
			low 6	1 mark		
			for b0 = 1 to 5	1 mark (including *next b0* below)		
			gosub wiper	(mark awarded above)		
			next b0	(mark awarded above)		
			goto main	(mark awarded above)		
						(17)

Q3:

Question	Mark Allocation	Marks	
7	 1 mark for each command with correct connections 1 mark for correct loop back to beginning	14 1 1	**16**
			(16)

Q4:

Question			Expected response	Max mark	Additional guidance
10.	(b)			8	1 mark – Motor On 4 marks – Decision box with correct Y/N directions (1 mark each decision) 1 mark Piston Out/Wait 1 s/In 1 mark Light On/Wait/Off 1 mark for light flashing in correct ratio (wait/delay 0·25 s) NB If a candidate uses the wrong symbol type then no mark, if they repeat the error for the same symbol allow FTE.

Stepper motor questions

No answers yet as stepper motors are not part of higher course as of 2019

A

A system is required to make a stepper motor make one eighth of a revolution then pause for 1 second before commencing the next turn. This should be repeated eight times after a start switch and a safety switch have been pressed. The motor moves in 7.5° per step and uses a step rate of 0.125 seconds (i.e. maximum of eight pulses per second). Draw a flow chart for this sequence (8 marks) and complete a PBASIC program using the following code (12 marks).

```
symbol direction = 7
symbol pulse = 6
symbol counter = b0
symbol counter2 = b1

let dirs = %11000000
let pins = 0
let direction = 0

main:
```

B

A system is required to make a motor turn for 10 seconds then pause for 1 second before turning in reverse for 10seconds. This should be repeated four times. A switch must be pressed to start the sequence. Draw a flow chart (12 marks) for this sequence and complete a PBASIC program on the right (8 Marks).

```
symbol counter = b0

let dirs = %11000000

main:
```